

CONFIDENTIAL

KAYAN
— DATA ARCHITECTURE —

Data Model & Schema

نموذج البيانات والسكيما

Every entity, every relation, every constraint

كل كيان وكل علاقة وكل قيد

KAY-TEC-003 · Rev 1

CONNECTING EXCELLENCE · EMPOWERING GROWTH

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EXPERT-REVIEWED CANONICAL DOCUMENT

Batch 13 — Technology Operating System · Reviewed 31 May 2026 · Reviewer: Claude (expert collective) · Ref: BATCH-13_Technology_Review-Memo.md

KAYAN OPERATING SYSTEM

Kayan Innovation, Entrepreneurship & Work Environments LLC

شركة كيان للابتكار وريادة الأعمال وبيئات العمل ذ.م.م

Data Model & Schema

نموذج البيانات والسكيما

Document code	Version	Effective date	Status
KAY-TEC-003	Rev 1	30 June 2026	ACTIVE on signature

Classification: INTERNAL. Document owner: Lead Developer (Mustafa Haney); CEO final authority on architecture; second developer Alaa. *Connecting Excellence, Empowering Growth!*

1. Status and authority

Canonical within the Operating System. Tech decisions reconcile here. Facts reconcile to KAY-GOV-LF-001; data and security implementation reconcile to KAY-COM-005 (Data Protection SOP). Mustafa Haney is cost-shared 50/50 with CauseWay under KAY-FIN-002; Alaa is Kayan-dedicated. Both paid EGP 18,000/month via InstaPay.

2. The core entities

Entity	Description	PII
Talent	A freelancer in the Trust Ladder. Identity, tier, screening, skills, engagements	Yes
Client	An institutional client (INGO, government, commercial, etc.)	Yes (signatory)
Engagement	A specific Managed Delivery engagement with its gate state, pod, deliverables	Some
SOW	Statement of Work — scope, criteria, pricing, dates	Some
Pod	The set of talents assigned to an engagement, with roles	No
Member	A Hub member with their product, payment status, access	Yes
Booking	A meeting-room booking by member, time, room, rate	No
Cohort	A Forge cohort with curriculum, participants, sponsor, dates	Some
Sponsor	A Forge sponsor with their cohort sponsorships	Yes (signatory)
Vendor	A supplier with their agreement, deliveries, payments	Yes (signatory)
GateEvent	A gate transition with timestamp, approver, evidence reference	No
QAReview	A Layer-1/2/3 QA review with reviewer, deliverable, outcome	No
Payment	A payment movement: receipt, talent payment, holdback, vendor	Some (account)

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IncidentReport	A logged incident per KAY-COM-007	Per type
DecisionLog	A logged decision per KAY-OPS-002	No

3. Relationships (key)

Engagement belongs-to Client and has-one SOW and has-one Pod. Pod has-many Talents (with role and tier-at-time-of-assignment). Engagement has-many GateEvents (one per transition). Engagement has-many QAReviews at G6 (typically three: self, pod-lead, independent). Talent has-many Engagements through Pod-membership. Cohort has-many Participants (Talents who later progress through verification). Sponsor has-many Cohort-sponsorships. Member is independent of Talent (most members are not talents; some are) — linked optionally where the same person is both. Payment links to Engagement (for talent payment + holdback) and to Vendor (for procurement) and to Member (for monthly billing) and to Sponsor (for cohort sponsorship).

4. Sensitive-data segregation

PII and financial data are stored in a separate restricted schema (talent_sensitive, client_sensitive, vendor_sensitive). The primary application database holds references and operational data; the restricted store holds: ID document images, passport scans, proof-of-address, bank-account confirmations, full bank-account numbers. Access to the restricted store is through specific named functions (get_kyc_documents, get_bank_account) audited at every call. The Lead Developer has technical access for maintenance; no business reading; access logged.

5. Audit log architecture

Every write operation on consequential entities (Talent tier, Engagement gate state, Payment, Sponsor agreement, Member status, Incident, Decision) writes an audit log entry: timestamp, actor (user ID), action, entity ID, before-state hash, after-state hash, source IP, user agent. The audit log is append-only (no update or delete); held in a partitioned table for performance; retained ten years per KAY-COM-006. Audit log queries are role-restricted; arbitrary reads are not permitted.

6. Soft-delete and retention

Records are not hard-deleted from the primary tables. "Deletion" sets a deleted_at timestamp; the record is filtered from standard queries but remains in the database for the ten-year retention window. After ten years, scheduled archival jobs export the record to long-term archive storage and remove from the primary database. Restricted store (sensitive PII) follows the same pattern with the additional protection that deletion explicitly clears sensitive fields after the retention period.

7. Schema versioning

Database migrations live in the repository, numbered and dated. Every migration is reviewed by both developers before merge. Forward-only migrations only; we do not rely on rollback in production. Schema changes touching restricted-store tables require CEO sign-off. Migrations run on staging before production; production runs are scheduled with notice; backups taken immediately before.

8. Backups and recovery

Daily backup of the primary PostgreSQL database to encrypted object storage in a different region. Weekly full backup retained for one year; monthly full backup retained for ten years. Restored to staging quarterly to verify

recoverability. Restricted-store has its own backup pipeline with stricter access controls. Backup encryption keys held in a key-management service with quorum unlock (CEO + Lead Developer).

9. Sample-data and test environments

Staging environment uses synthetic test data, never production data. Where production-shape data is needed for testing (a specific bug), the developer uses anonymised production snapshots — PII fields scrubbed, identifiers randomised — generated through a documented anonymisation function. Production database is never copied to a developer workstation. Read access to production for live debugging is logged and time-bound.

Authority and change control

Amended only by signed revision with Master Review Log entry.

Signatures and authorisation — التوقيعات والاعتماد

Role / الدور	Name / الاسم	Signature / التوقيع	Date / التاريخ
CEO & Founder — المدير التنفيذي والمؤسس	Maher Faisal Saeed Farea	_____	___/___/2026
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